

MTE 203 Week 1 Tutorial

Trim Ch 11, S. 11.5, 11.6



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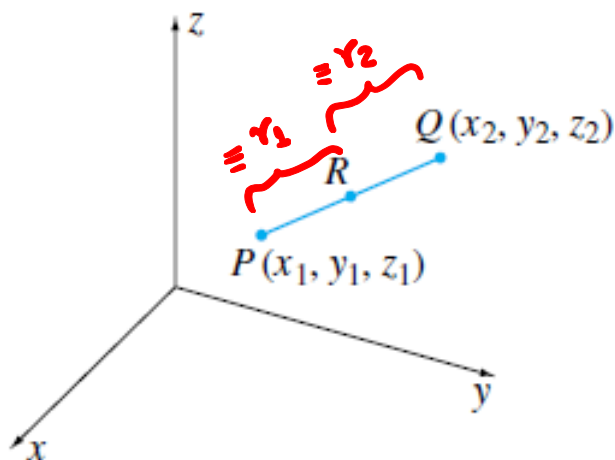
Problem 1: (Ch11.1, Prob 19)

If P and Q in the figure below have coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) , prove that if a point R divides the length of the segment PQ so that,

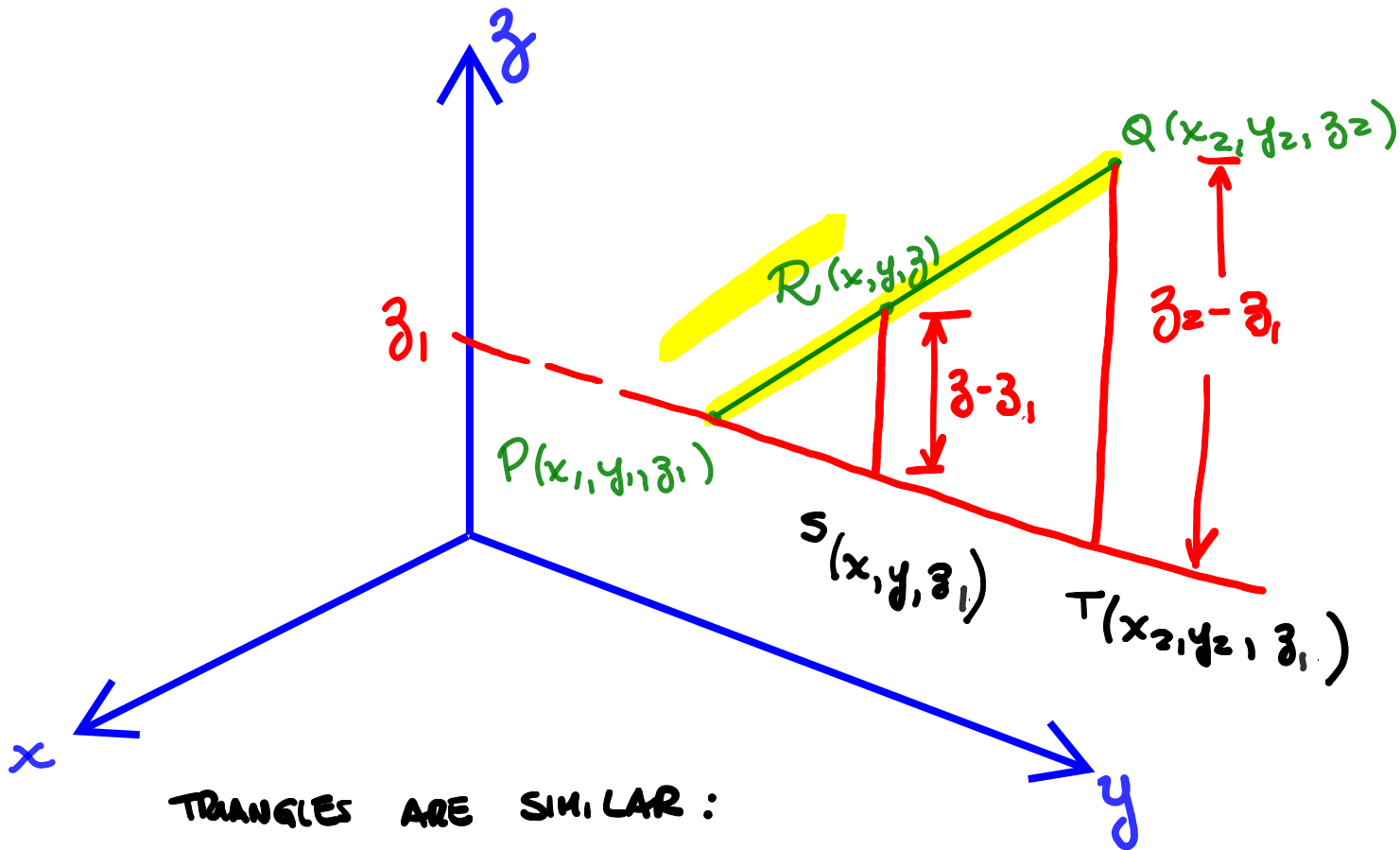
$$\frac{|\overrightarrow{PR}|}{|\overrightarrow{RQ}|} = \frac{r_1}{r_2},$$

where r_1 and r_2 are positive integers, then the coordinates of R are,

$$x = \frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \quad y = \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2}, \quad z = \frac{r_1 z_2 + r_2 z_1}{r_1 + r_2}$$



Problem 1 - Solution



TRIANGLES ARE SIMILAR :

$$\frac{|\vec{PQ}|}{|\vec{PR}|} = \frac{|\vec{QT}|}{|\vec{RS}|} = \frac{z_2 - z_1}{z - z_1} \dots (1)$$



Problem 1 - Solution

USING (1) AND SUBTRACTING "-1"

$$\frac{|\vec{PQ}|}{|\vec{PZ}|} - 1 = \frac{z_2 - z_1}{z - z_1} - 1$$

$$\frac{|\vec{PQ}| - |\vec{PZ}|}{|\vec{PZ}|} = \frac{z_2 - \cancel{z_1} - z + \cancel{z_1}}{z - z_1} = \frac{r_2}{r_1}$$

$$\Rightarrow z = \frac{r_1(z_2 - z)}{r_2} = \frac{r_1 z_2 + r_2 z_1}{r_1 + r_2}$$

FOR MIDPOINT

$$r_1 = r_2 \Rightarrow z = \frac{r z_2 + r z_1}{r + r} = \frac{z_2 + z_1}{2}$$

FIND x AND $y \Rightarrow$ FOR YOU 😊



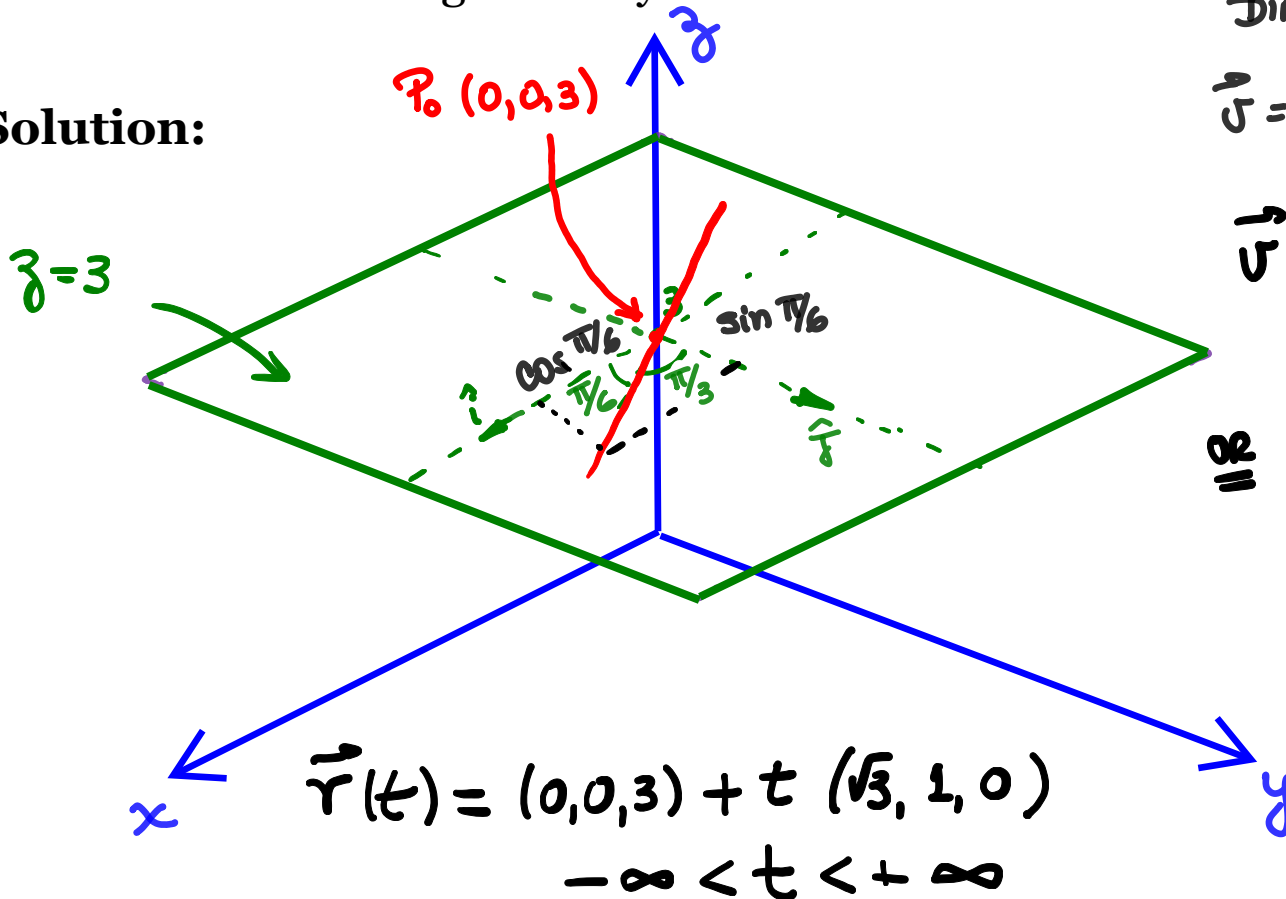
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Problem 2

Find the vector form of the equation for the line in the plane $z = 3$ than makes an angle of $\frac{\pi}{6}$ rad with $\hat{i}$ and of $\frac{\pi}{3}$ rad with $\hat{j}$.

Describe the reasoning behind your answer.

Solution:



DIRECTION OF LINE

$$\vec{v} = (\cos \frac{\pi}{6}, \sin \frac{\pi}{6}, 0)$$

$$\vec{v} = (\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$$

$$\parallel \vec{v} = (\sqrt{3}, 1, 0)$$

$$P_0: (0,0,3)$$



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Problem 3

The lines L_1 and L_2 are represented by the following parametric equations,

$$L_1: \quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$

$$L_2: \quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s$$

- a) Write their vector form
- b) Find their symmetric equations
- c) Show that L_1 and L_2 do not intersect

Solution:

$$L_1: \quad \vec{r}(t) = (1, -2, 4) + t \underline{(1, 3, -1)}, \quad -\infty < t < \infty$$

$$L_2: \quad \vec{r}(s) = (0, 3, -3) + \underline{s(2, 1, 4)}, \quad -\infty < s < \infty$$



Problem 3 - Solution

$$L_1: x-1 = \frac{y+2}{3} = 4-z, -\infty < t < \infty$$

$$L_2: \frac{x}{2} = y-3 = \frac{z+3}{4}, -\infty < s < \infty$$

C) IF THEY INTERSECT, THERE EXISTS A "t" AND "s" THAT SATISFY THE FOLLOWING EQUATIONS,

$$\begin{array}{l} \text{FROM} \\ \text{PARAMETRIC} \\ \text{EQ'S} \end{array} \left\{ \begin{array}{l} 1+t = 2s \quad \dots (1) \\ -2+3t = 3+s \quad \dots (2) \\ 4-t = -3+4s \quad \dots (3) \end{array} \right.$$



Problem 3 - Solution

FROM (1) AND (2):

$$\begin{array}{r} 1+t = 2s \\ -4+6t = 6+2s \\ \hline -5+5t = 6 \\ t = \frac{11}{5} \rightarrow s = \frac{8}{5} \end{array}$$

REPLACING IN (3):

$$\underbrace{4 - \frac{11}{5}}_{\frac{9}{5}} \neq \underbrace{-3 + 4\left(\frac{8}{5}\right)}_{\frac{17}{5}} \Rightarrow \text{THEY DO NOT INTERSECT !!}$$

THEY ARE NOT PARALLEL AND DO NOT INTERSECT
 $\rightarrow$ SKEWED LINES



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Thank You